Repair Bias in Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected coverage-repair record.

Locate missing groups, combine counts, broaden collection and challenge causal shortcuts.

Keep one habit: Keep group numerators, denominators, collection methods, proposed changes and unobserved outcomes distinct.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Sampling and Representation

Noise and Outliers

Decision Trees

W1 · Locate missing visitor coverage

> State which time groups the invitation process excludes.

Your evidence record

- Opening sample: club8 Yes/10 responses, other2/10; combined counts, not an unweighted average, define the observed sample rate. A separate90/10 example has72 club Yes and2 other Yes. The pictured comic poll instead has12/15 and3/15, grade30 club/90 other, and volunteer sampling. Population representativeness is not established by arithmetic or weights alone.

- A morning-only survey claims to represent all library visitors. Which visitors may be missed?

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Combine the supplied3/4 and2/6 counts

> Use total Yes divided by total respondents.

Your evidence record

- Opening sample: club8 Yes/10 responses, other2/10; combined counts, not an unweighted average, define the observed sample rate. A separate90/10 example has72 club Yes and2 other Yes. The pictured comic poll instead has12/15 and3/15, grade30 club/90 other, and volunteer sampling. Population representativeness is not established by arithmetic or weights alone.

- Group A has 3 Yes of 4 and B has 2 Yes of 6. Find the combined proportion.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair a restricted collection process

> Describe a broader frame, access and nonresponse record.

Your evidence record

- Opening sample: club8 Yes/10 responses, other2/10; combined counts, not an unweighted average, define the observed sample rate. A separate90/10 example has72 club Yes and2 other Yes. The pictured comic poll instead has12/15 and3/15, grade30 club/90 other, and volunteer sampling. Population representativeness is not established by arithmetic or weights alone.

- A form is shared only in the robotics club. How can the **process** better cover the grade?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Limit a causal claim

> Preference may influence membership rather than the reverse.

Your evidence record

- Opening sample: club8 Yes/10 responses, other2/10; combined counts, not an unweighted average, define the observed sample rate. A separate90/10 example has72 club Yes and2 other Yes. The pictured comic poll instead has12/15 and3/15, grade30 club/90 other, and volunteer sampling. Population representativeness is not established by arithmetic or weights alone.

- Club members favor coding more often. Does club membership necessarily cause that preference?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Audit independently

> Keep one first attempt before hints or feedback.

Read and explain independently

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Audit independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Change feature coverage or weights only after locating which reader profiles are underserved.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Combine counts for **Selection bias** and coverage. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Selection bias** and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

Propose repair and evaluation.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A dataset has 18 adventure readers and 2 poetry readers. A recommender ignores poetry preferences. What would you inspect and test after a repair?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 20 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

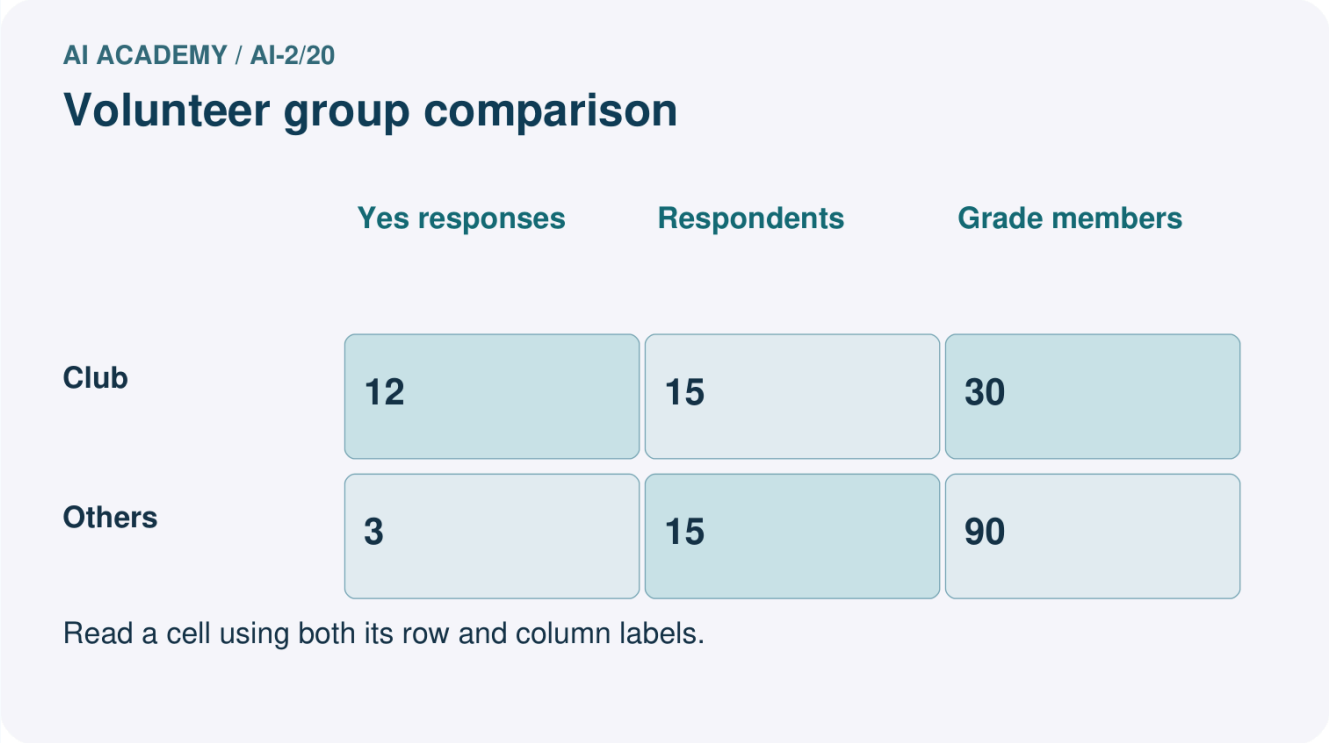
> Keep the explanation and the support you used.

Use this when helpful

A fictional school polls learners about adding comic recommendations.

Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15.

The full grade has 30 club members and 90 others. These respondents were volunteers, not random selections.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/20

Repair Bias in Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Group rates

Club:12/15=80%; others:3/15=20%.

↓

2. Respondent total

15 Yes out of 30 respondents =50%.

↓

3. Check representation

Respondents split 50/50, but the grade is 25% club and 75% other.

↓

4. Repair collection

Plan invitations that fairly cover the whole grade, document selection and report nonresponse by group.

CHANGE ONE THING

Recruit more club volunteers only.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional school polls learners about adding comic recommendations.
- Which labelled step matches this detail? Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.
- 1. Group rates
- 2. Respondent total
- 3. Repair collection
- 4. Check representation
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Repair collection Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.

The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact.

Changed-case question

- Changed-case practice: Recruit more club volunteers only. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Sample size increases without fixing missing representation of other learners.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Sample size increases without fixing missing representation of other learners. The added respondents come from the already overrepresented group.

Repair and save your work

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

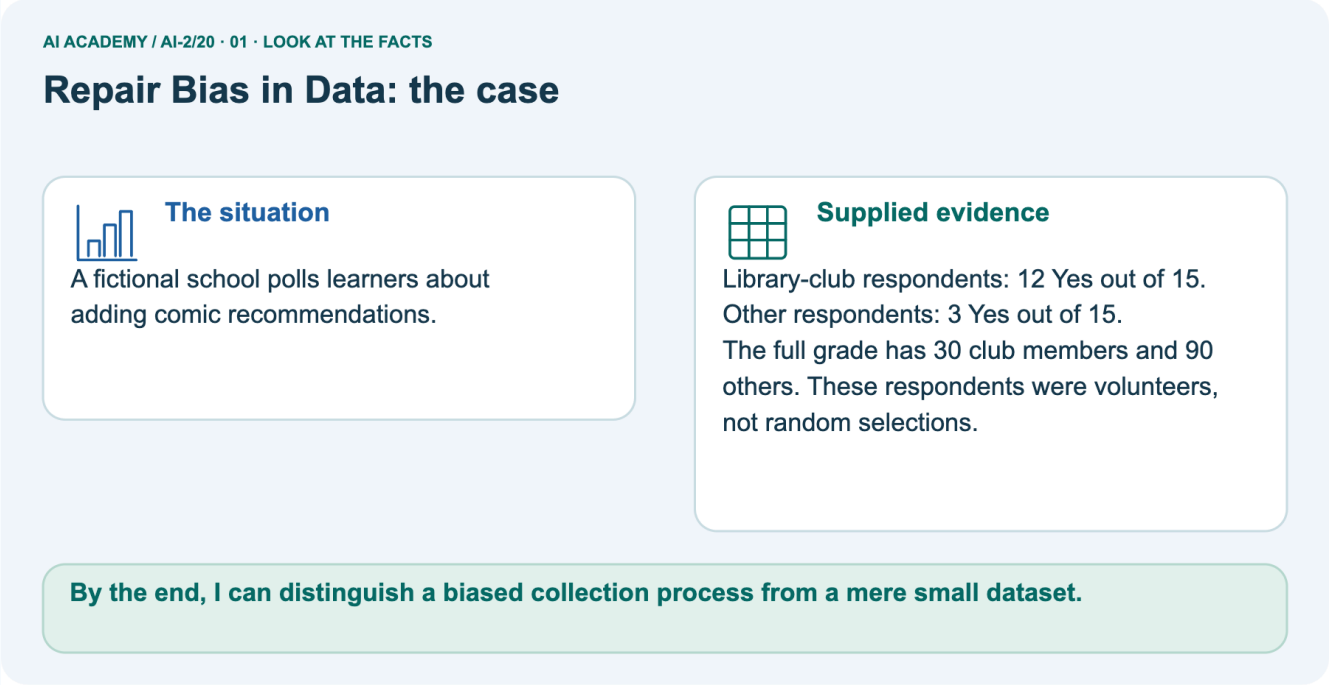
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Repair Bias in Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

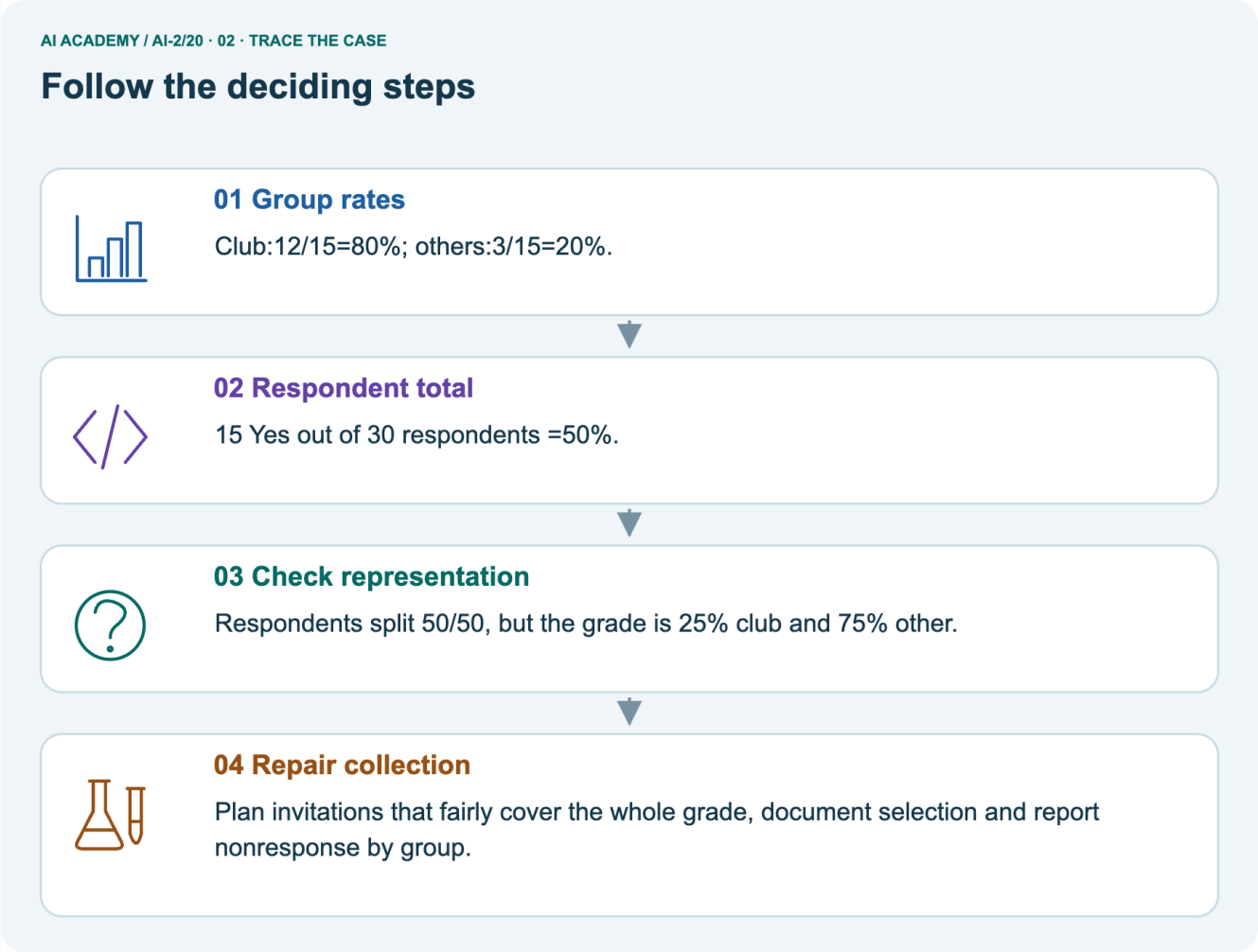
A fictional school polls learners about adding comic recommendations. Supplied facts: Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15. The full grade has 30 club members and 90 others.

These respondents were volunteers, not random selections.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



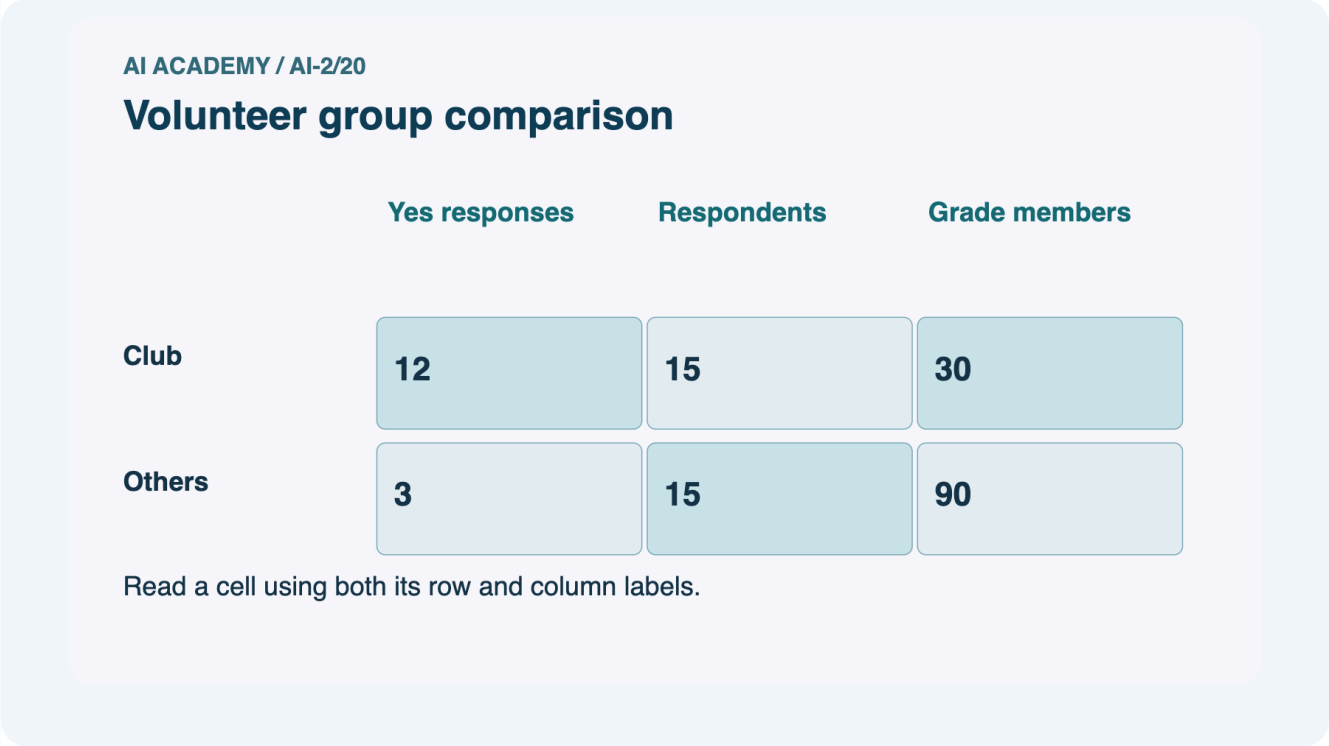
Follow the labelled picture with the matching explanation. Open the vector diagram

Group rates: Club:12/15=**80%**; others:3/15=20%. → Respondent total: 15 Yes out of 30 respondents =50%. → Check representation: Respondents split 50/50, but the grade is 25% club and 75% other. → Repair collection: Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Volunteer group comparison



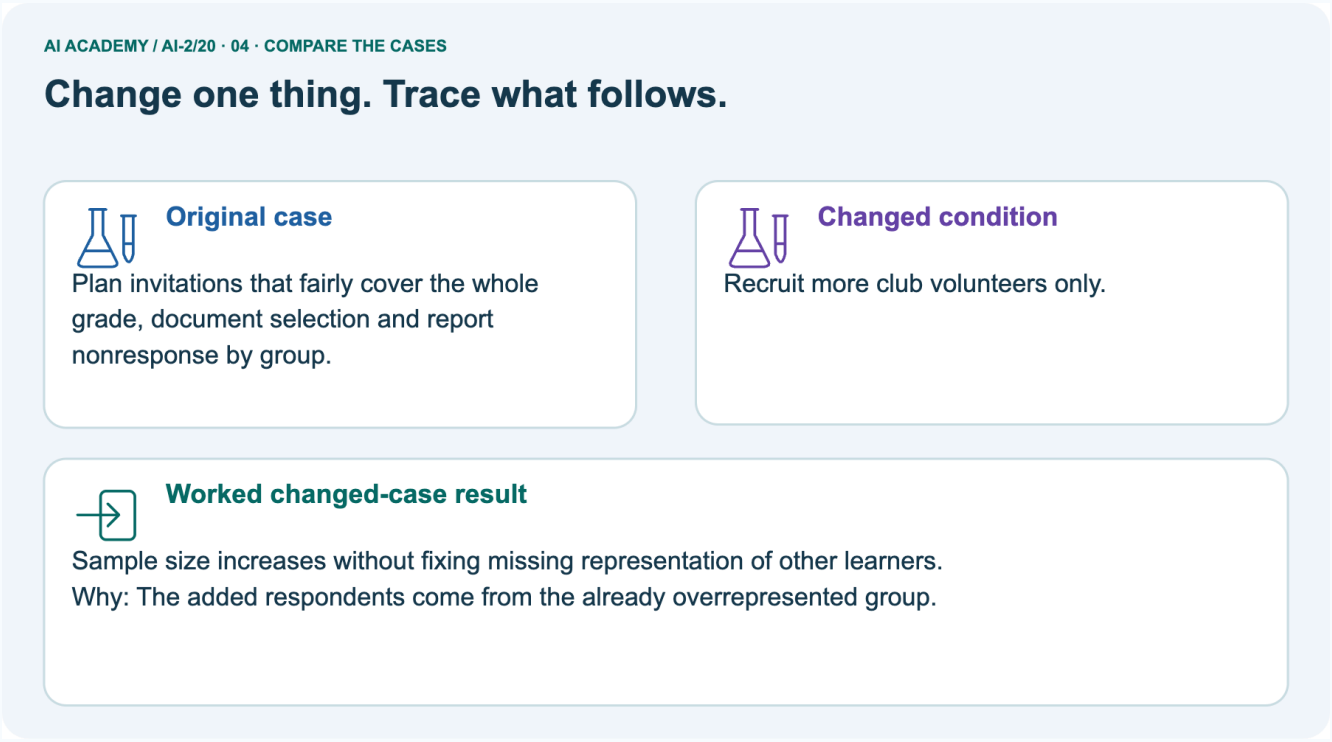
Follow the labelled picture with the matching explanation. Open the vector diagram

Volunteer group comparison. Club: Yes responses = 12, Respondents = 15, Grade members = 30. Others: Yes responses = 3, Respondents = 15, Grade members = 90.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

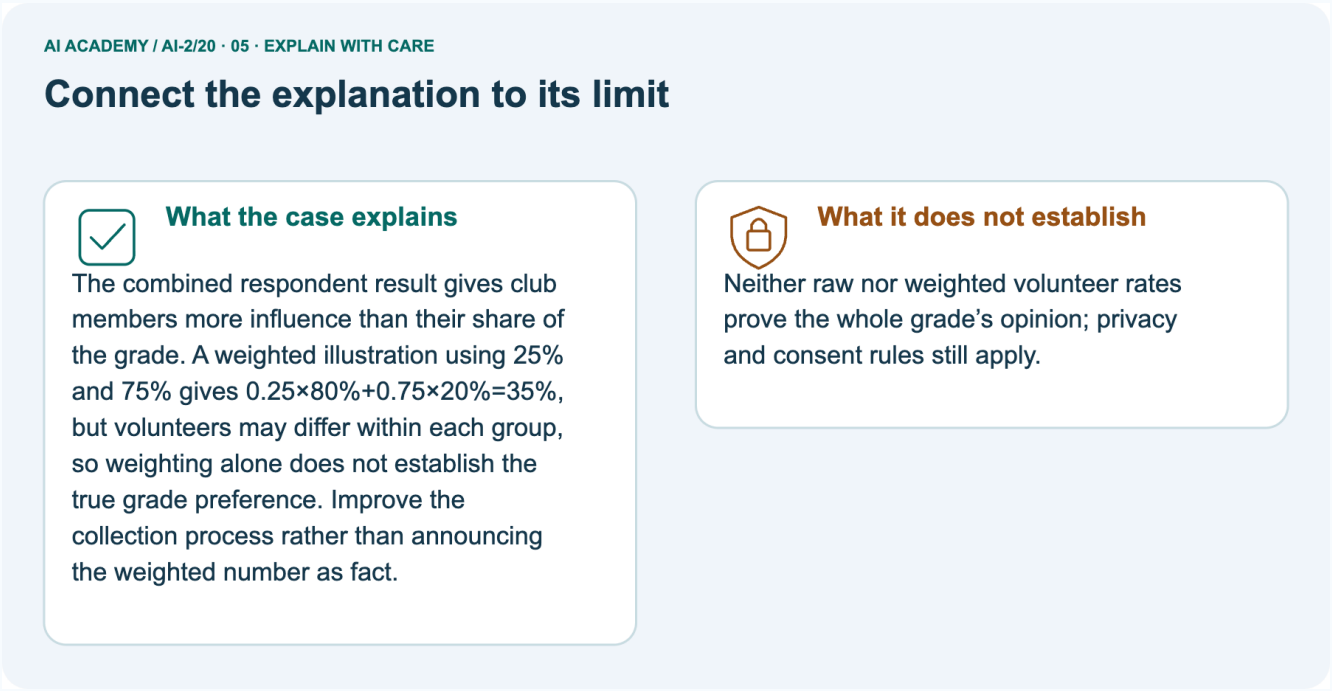
Original case: Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group. Changed condition: Recruit more club volunteers only.

Expected result: Sample size increases without fixing missing representation of other learners. Reason: The added respondents come from the already overrepresented group.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The combined respondent result gives club members more influence than their share of the grade.

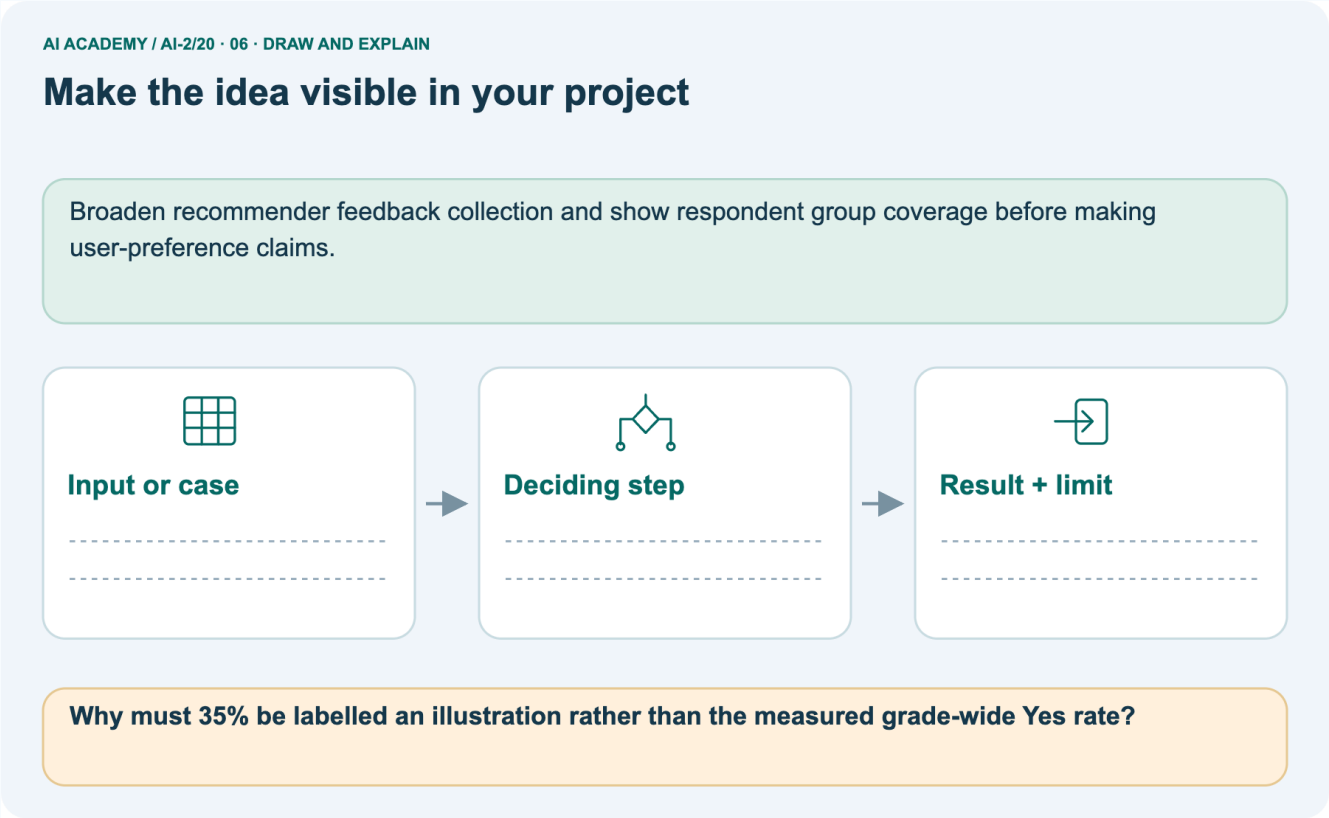
A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact. Limit: Neither raw nor weighted volunteer rates prove the whole grade's opinion; privacy and consent rules still apply.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims. Draw your **input**, deciding step, result and limit.

Supported question: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Repair Bias in Data: the case

02. Follow the deciding steps

03. Volunteer group comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.